HDI PCB TECHNOLOGY Enhancing The Performance Of Equipment

High-density interconnection (HDI) PCB takes advantage of recent technologies to amplify the functionality of circuit boards, motivated by the tininess of parts and semiconductor packages that provide superior characteristics in innovative new products, including touchscreen phones

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The first printed circuit board (PCB) can be traced all the way back to the early 1940s. Back then, the US needed basic PCB for military applications. Since then, PCB technology has constantly evolved, from single-sided, double-sided to multilayer, flex, rigid-flex and so on.

For many years, PCB design was dictated by the evolution of components in the quest to provide more efficiency, more capabilities and bigger connection density. The trend is still the same; components continue to decide the evolution of PCB design. Power LEDs are a good example of this.

The same goes for high-density packaging components like ball-grid array (BGA) and micro-BGA. PCB designers have pushed PCB manufacturers to build circuits capable of supporting such fine pitch components.

Engineers, circuit board manufacturers, raw material suppliers and PCB manufacturers have developed processes and laminates to print the design for high-density components. By reducing the pitch, interconnection is getting more and more complex. Vias/holes have to be reduced because pads space has reduced, too.

What is an HDI PCB

HDI stands for high-density interconnection. A circuit board with a higher wiring density per unit area as opposed to a conventional board is called HDI PCB. An HDI PCB has finer spaces and lines, minor vias and capture pads, and higher connection pad density. HDI is the result of component packaging evolution. Most of the new components need PCBs with high density for enhancing electrical performance.

HDI boards are one of the fastest growing technologies in PCB. These contain blind and/or buried vias, and often contain micro-vias that are 0.006 or less in diameter. These have a higher circuitry density than traditional circuit boards.

Now, products do more, weigh less and are physically smaller. Speciality equipment, mini-components and thinner materials have allowed for electronics to shrink in size while expanding in technology, quality and speed.

There are different HDI structures:

x + N + x

Here, first x is the number of added layers on the top and bottom sides, N is the initial built-up multilayer (4, 6, 8, … layers) and second x is the number of added layers on the top side.

Fig. 1: Structure of HDI PCB